Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	Kebanggan Universiti Malaya (KUM)
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 8 — Decent Work and Economic Growth, SDG 9 Industry, Innovation and Infrastructure
Project / Solution Name	Dispatch Hub
Name (Leader)	Tan Kai Xue
Email & Phone (Leader)	duopoitan@gmail.com / 0167158480
University Affiliation(s)	Universiti Malaya
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysian micro and small service businesses, such as air-conditioning, electrical, plumbing, appliance and equipment repair teams, run their day on chat apps, spreadsheets and memory. When a customer reports a fault, the owner becomes the control center, deciding who to send, messaging technicians one by one, waiting for someone to say "OK" and chasing again when nobody does. It is a recurring burden that keeps owners working in the evening and still produces drop jobs and slow response.

Dispatch Hub is our answer. It is an automation product for the Orbyte ecosystem that runs the **full line of a work order**. A manager approves the distribution, the technician accepts it in the website, every step is logged, and AI turns the log into practical suggestions for improvement.

We address one main challenge—distributing and accepting work orders—by creating a complete, easy-to-follow workflow. It guides the process from an initial trigger, through data collection and human approval, right to a visible outcome. Most importantly, it's a reusable design that other Orbyte SMEs can easily apply to their own work.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

We use ai to read the activity log, including time to accept, decline reasons, offers no response, busy areas and time slots, repeat faults and parts shortages. The ai will produce a short weekly summary in plain language with specific suggestions, each linked to its evidence. The manager decides whether to act on or dismiss each suggestion.

For example: "This week 3 of 9 offers sent between 8 and 10 am were declined, mostly as 'already on another job'. Consider staggering morning jobs or keeping one technician on standby." The figures are calculated from the log by code, and the AI only writes the explanation and suggestions, so numbers cannot be invented. Dispatch decisions with rules and people.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Layer	Technology
Frontend	React, TypeScript, Vite, Tailwind CSS
Backend	Node.js 24, Express 5, TypeScript
Database	SQLite with `better-sqlite3` and Drizzle tools
Validation	Zod and generated API schemas/hooks
API contract	OpenAPI and Orval-generated client code
AI / Compute	Optional RunPod Serverless endpoint
Maps	Leaflet/OpenStreetMap display and optional Google Routes
Hosting / Dev	Replit, pnpm workspaces

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

The prototype connects a web interface to a backend API, with RunPod used for selected AI tasks and Google Maps used for navigation. The AI is constrained by server-side rules and output validation; some diagnosis and technician-recommendation flows have rule-based fallbacks. The design also keeps consequential decisions, such as dispatch approval and technician acceptance, under human control.

These choices demonstrate a feasible workflow, but they do not by themselves establish production readiness. Phase 2 should include role-based authentication and access controls, privacy and retention decisions for information sent to AI services, testing with intended users and operational data, model-quality and fairness evaluation, monitoring and error handling for external services, and deployment and recovery testing. Until those checks are complete, the prototype should be treated as a demonstration rather than a production-ready service.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess **Innovation and Creativity** as well as **Feasibility of build**. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 11.6: Reduce the adverse environmental impact of cities, including by paying special attention to air quality. The project potential contribution is to reduce avoidable service-van return trips, not optimizing routes or directly measuring air quality.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	0% of work orders have a verified environmental outcome recorded in the prototype. The prototype does not currently track first-visit completion, repeat-trip reasons, fuel use, or emissions. This means there is no measured environmental reduction to claim yet; it does not mean the real-world repeat-trip rate is zero.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Preventable repeat visits per 100 completed work orders, where the return trip was caused by an incorrect initial job assessment, unsuitable technician assignment, or missing parts. The product's AI job analysis and technician/van stock matching are intended to help reduce these cases.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation*	Reduce preventable repeat visits by 10% against a baseline measured before using the product, during a pilot or controlled test. Use that reduction to estimate the

Metric Category	Your Response
e.g., "20% improvement in computational resource use."	corresponding decrease in tailpipe pollutants that affect air quality, such as NOx and PM2.5

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [x] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [x] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [x] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [x] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React, Vite, Express, SQLite (better-sqlite3), Drizzle ORM, Zod, Tailwind CSS, TanStack Query, React Hook Form, Radix UI, Lucide, and other UI and utility packages.	Open-source frameworks and libraries	
RunPod endpoint	Pre-trained AI	
Google Maps Platform	Third-party APIs	

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[25 October 2026]